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Locking mechanism and container for delivering items

Abstract

A canister is coupled to a delivery vehicle using a rail bracket. The canister comprises a securing channel having a first securing channel volume and a second securing channel volume. A width of the first securing channel volume is less than a width of the second securing channel volume. The rail bracket comprises a rotational shaft having a finger at one end. The rail bracket is inserted into the securing channel, and the rotational shaft is rotated. When rotated, the finger inhibits removal of the rail bracket from the securing channel, thereby providing a mechanism by which the canister can be releasably coupled to the rail bracket. The rail bracket may be provided on a delivery vehicle, such as an unmanned aerial vehicle (UAV) to transport the canister and contents within it.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application U.S. patent application Ser. No. 17/818,848, entitled “Locking Mechanism and Container for Delivering Items,” filed on Aug. 10, 2022; which claims priority to U.S. Provisional Application No. 63/237,392, entitled “Pneumatic Platform and Payload Containers for Unmanned Aerial Vehicles,” filed on Aug. 26, 2021, each of which is hereby expressly incorporated by reference in its entirety.

BACKGROUND

(1) Unmanned aerial vehicles (UAVs) are used to deliver packages. In doing so, a UAV retrieves a package and releases it at a delivery location.

SUMMARY

(2) At a high level, aspects described herein relate to a canister and a rail bracket for releasably coupling the canister to a UAV, or other delivery vehicle.

(3) An aspect of the canister comprises a first canister end that is opposite from a second canister end. The first canister end and the second canister end define a longitudinal direction. The canister comprises a wall that extends between the first canister end and the second canister end, and the

wall defines an interior volume of the canister. Objects for delivery can be placed within the interior volume. The canister has a first securing channel with a first channel volume and a second channel volume. The first channel volume has a first width that is less than a second width of the second channel volume when the first and second widths are measured in the longitudinal direction. The second channel volume is more interior to a center aspect of the interior volume than the first channel volume.

(4) An aspect of the rail bracket comprises a first bracket surface opposite a second bracket surface, such that the first bracket surface is exposed to the UAV. The rail bracket further comprises a first rotational shaft having a first shaft end and a second shaft end. The first rotational shaft extends through the first bracket surface and the second bracket surface, and comprises a first rotational axis. The first rotational shaft further comprises a first finger that extends radially from the first rotational shaft proximate to the second shaft end. The first rotational shaft extends a first radial distance from the first rotational axis, and the first radial distance is less than a second radial distance over which the first finger extends from the first rotational axis.

(5) The rail bracket releasably couples the canister to the UAV. To do so, the rail bracket is inserted at least partially into the first securing channel, and the first rotational shaft is rotated, thereby transitioning the first finger from a first position to a second position, where the first position is substantially perpendicular to the longitudinal direction, and the second position is offset from the first position. In the first position, the canister is decoupled from the UAV, while in the second position, the canister is coupled to the UAV.

(6) This summary is intended to introduce a selection of concepts in a simplified form that are further described below in the detailed description section of this disclosure. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be an aid in determining the scope of the claimed subject matter. Additional objects, advantages, and novel features of the technology will be set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the following or learned by practice of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present technology is described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1 is an example UAV comprising an example rail bracket, in accordance with an aspect described herein;

(3) FIG. 2 is an example canister suitable for use with the rail bracket and UAV of FIG. 1, in accordance with an aspect described herein;

(4) FIG. 3A is a longitudinal cross section of the canister of FIG. 2, in accordance with an aspect described herein;

(5) FIG. 3B is an expanded view of a portion of the longitudinal cross section of FIG. 3 that includes a first securing channel, in accordance with an aspect described herein;

(6) FIG. 3C is an expanded view of a portion of the longitudinal cross section of FIG. 3 that includes a second securing channel, in accordance with an aspect described herein;

(7) FIG. 4 is an example rail bracket suitable for use with the UAV of FIG. 1 and the canister of FIG. 2, in accordance with an aspect described herein;

(8) FIG. 5A is a cross sectional view of a portion of the rail bracket of FIG. 4 as inserted into the first securing channel of the canister of FIG. 2, where the first finger is in a first position, in accordance with an aspect described herein;

(9) FIG. 5B is a cross sectional view of a portion of the rail bracket of FIG. 4 as inserted into the

first securing channel of the canister of FIG. 2, where the first finger is in a second position, in accordance with an aspect described herein;

(10) FIG. 6 is an example UAV comprising an example canister coupled thereto, in accordance with an aspect described herein;

(11) FIG. 7 is the UAV of FIG. 6 comprising the canister decoupled from the UAV, in accordance with an aspect described herein; and

(12) FIG. 8 is another example securing channel having a series of grooves that is suitable for use with any of the aspects described in FIGS. 1-7.

DETAILED DESCRIPTION

(13) UAVs are useful in delivering items because there are fewer restrictions on travel paths, making last-mile delivery more efficient in some cases. Some UAVs house items within a fuselage, while some grasp items using external mechanisms. A goal of UAV delivery is for the UAV to retrieve the item at one location and to release the item at a second location.

(14) The present disclosure provides a canister and a rail bracket that can be used in cooperation to deliver items. The canister includes an interior volume where items for delivery can be placed. The canister is releasably secured to the UAV by way of the rail bracket. Thus, an item can be loaded into the canister, and the canister secured to the UAV. The UAV navigates from one location to another location where it releases the canister, thereby delivering the object within it.

(15) One example canister that can be used to accomplish this task comprises a first canister end that is opposite a second canister end. The canister includes a wall that at least partially defines the interior volume between the first canister end and the second canister end. The canister extends from the first canister end to the second canister end in a longitudinal direction.

(16) To facilitate coupling to the UAV, the canister has one or more securing channels, including a first securing channel. The first securing channel has a first channel volume and a second channel volume. The first channel volume has a first width that is less than a second width of the second channel volume in the longitudinal direction. Further, the second channel volume is more interior to a center aspect of the interior volume than the first channel volume.

(17) The rail bracket is used in cooperation with the canister. One example of a rail bracket comprises a first bracket surface opposite a second bracket surface. When in use with a delivery vehicle, such as a UAV, the first bracket surface is exposed to the delivery vehicle.

(18) The rail bracket further comprises one or more rotational shafts, including a first rotational shaft. The first rotational shaft has a first shaft end opposite from and extending to a second shaft end. A first rotational axis extends through the first shaft end and the second shaft end. The first rotational shaft extends through the first bracket surface and the second bracket surface. Moreover, the first rotational shaft extends outward from the first rotational axis over a first radial distance.

(19) The first rotational shaft has a first finger that is proximate the second shaft end. The first finger extends radially from the first rotational shaft over a second radial distance. The first radial distance is less than a second radial distance.

(20) The rail bracket can be used to releasably couple the canister to the UAV such that the canister can be coupled and decoupled through rotation of the first rotational shaft. Here, the rail bracket is inserted at least partially into the first securing channel, and the first rotational shaft is rotated, such that it transitions the first finger from a first position to a second position. In the first position, the first finger is able to pass through the first channel volume and into the second channel volume. When rotated, the second position of the first finger is offset from the first position. The first finger rotates within the second channel volume. As an example, the first finger may be substantially perpendicular to the longitudinal direction when in the first position, and is then offset from the first position when rotated to the second position. In the second position, the first finger does not pass back through the first channel volume, thereby coupling, e.g., locking, the canister into place. Thus, in the first position, the canister is decoupled from the UAV, while in the second position, the canister is coupled to the UAV.

(21) This arrangement is particularly beneficial in that the force applied to the securing bracket by the mass of the canister is not transferred to the rotational device that rotates the first rotational shaft. Rather, the force is transferred to other mechanical elements of the UAV or rail bracket. This allows smaller devices that apply relatively less force, such as small rotary solenoids, to be used, thereby reducing overall weight of the system and wear on the device that is used to release and secure the canister by rotating the first rotational shaft.

(22) The aforementioned embodiments have been provided as examples of the technology that may be practiced from the present disclosure. They are intended to be examples and are provided to aid in understanding the technology and its benefits. Additional features and embodiments are further described with reference to the figures.

(23) FIG. 1 illustrates example UAV **100** comprising example rail bracket **102**. While UAV **100** is illustrated as a four rotor UAV, the present technology may be employed with various other UAV systems, and UAV **100** is intended to be illustrative in nature. Such UAV systems include, but are not limited to, vertical takeoff and landing systems, forward flight wing-based systems, and combinations thereof, along with any autonomous, semi-autonomous, or piloted system, and other like systems.

(24) As illustrated, UAV **100** has coupled to it rail bracket **102**. In general, UAV **100** may comprise one or more rail brackets, such as rail bracket **102**. Any one or more rail brackets may be provided at any location, and in any arrangement, on a UAV to facilitate releasably coupling a canister to the UAV, as will be further described. In one particular position, such as that shown in FIG. 1, one or more rail brackets, such as rail bracket **102**, are positioned on an underside of a fuselage. In an embodiment, rail bracket **102** is aligned with a center of gravity of UAV **100**. As will be further described, alignment with the center of gravity may allow for additional power efficiencies when the UAV **100** is used to transport a canister using rail bracket **102**.

(25) FIG. 2 illustrates an example canister **200** that is suitable for use with UAV **100** and rail bracket **102** of FIG. 1. In the example illustrated, canister **200** comprises first canister end **202** that is opposite from second canister end **204**. Canister **200** extends from first canister end **202** to second canister end **204** along a longitudinal direction. Put another way, first canister end **202** is opposite second canister end **204** and defines the longitudinal direction therebetween.

(26) Canister **200** may comprise any length. To provide some examples, canister **200** may have a length in the longitudinal direction that is from about 5 inches (in.) (12.7 centimeters (cm)) to about 15 in. (38.1 cm). Canister **200** length may be selected based on an item that will be carried within canister **200** or based on a size of a delivery vehicle carrying canister **200**.

(27) Canister **200** further comprises wall **206**, which at least partially extends between first canister end **202** and second canister end **204**. Wall **206** may be formed of any rigid material, such as a rigid polymer, e.g., PVC (polyvinyl chloride), HDPE (high-density polyethylene), PEEK (polyetheretherketone); a metal or metal alloy, e.g., aluminum, stainless steel; or another like material. Wall **206** may define at least a portion of interior volume **208**. Interior volume **208** can correspond to an accessible chamber suitable for positioning objects therein, such as an item that is being delivered using canister **200**.

(28) Interior volume **208** comprises a center aspect **210**. In the example illustrated, canister **200** is generally cylindrical, e.g., it comprises a polygonal or circular cross section at at least one location perpendicular to and along the longitudinal direction. Thus, center aspect **210** extends along the longitudinal direction at a midpoint of a cross section of the cylinder. In other arrangements, center aspect **210** may be determined based on the geometric shape of the canister and its corresponding interior volume.

(29) Canister **200** comprises one or more securing channels. In general, a securing channel comprises a location or area at which a canister, such as canister **200**, is coupled to another object. In aspects, a canister may be coupled at a securing channel to a delivery vehicle that moves the canister from one location to another. One specific delivery vehicle with which a canister, such as

canister **200**, can be used is a UAV. Other delivery vehicles include ground based vehicles, such as cars, trucks, trailers, bicycles, and the like; conveyor systems, sorting systems, and the like; and aircraft. Such delivery systems may be manned, unmanned, autonomous, semi-autonomous, onboard operated, or remotely operated.

(30) In the example illustrated by FIG. 2, canister **200** comprises first securing channel **212** and second securing channel **214**. As noted, and in general, a canister may comprise any number of one or more securing channels. In the example illustrated, each of first securing channel **212** and second securing channel **214** extends at least partially around an outer aspect of canister **200**. In embodiments, a securing channel, such as first securing channel **212** and second securing channel **214**, extends entirely around an outer aspect of the canister.

(31) As previously noted, canister **200** is substantially cylindrical, e.g., it has a polygonal or circular cross section at at least one location along the longitudinal direction. In a particular embodiment, the cross section is perpendicular to the longitudinal direction. As will be appreciated, all or a portion of canister **200** maybe substantially cylindrical. Thus, in a particular aspect, the polygonal or circular cross section is at a location corresponding to first securing channel **212** or second securing channel **214**. In such cases, first securing channel **212** or second securing channel **214** may extend entirely or at least in part around the polygonal or circular cross section. In a particular aspect, first securing channel **212** or second securing channel **214** extends along a theoretical plane that is perpendicular to the longitudinal direction.

(32) It will be understood, that, while the example provided herein is a substantially cylindrical canister, such mechanisms may be used in coordination with other objects of any size or shape. For instance, a first and second securing channel, arranged as previously described, may extend about any object being secured with a rail bracket, as will be further described. For instance, first and second securing channel may extend about a substantially flat surface, such as on a shipping container. In some cases, the orientation of these objects, when retrieved or released, is fixed, and thus, rotation about the first and second securing channel, as will be described, may not be required. In these cases, a rail bracket system of a delivery vehicle may interact with the first and second securing channels in manners similar to those provided herein, in order to release or retrieve the object for delivery.

(33) As provided in the illustrated example of FIG. 2, first securing channel **212** is proximate first canister end **202**. Thus, first securing channel **212** may be positioned at first canister end **202** or separated from first canister end **202** by a distance measured in the longitudinal direction. For instance, first securing channel **212** may be positioned at first canister end **202** or anywhere between first canister end **202** and a theoretical midline of canister **200** measured in the longitudinal direction. In example aspects, a portion of first securing channel **212** may be positioned at any a location from about 0.25 in. (0.64 cm) to about 7.5 in. (19.1 cm) away from first canister end **202**, as measured in the longitudinal direction. In a particular embodiment not illustrated, first securing channel **212** is positioned at the theoretical midline, e.g., at a midpoint between first canister end **202** and second canister end **204**.

(34) In aspects that comprise second securing channel **214**, such as that illustrated in FIG. 2, second securing channel **214** is proximate second canister end **204**. Thus, second securing channel **214** may be positioned at second canister end **204** or separated from second canister end **204** by a distance measured in the longitudinal direction. For instance, second securing channel **214** may be positioned at second canister end **204** or anywhere between second canister end **204** and a theoretical midline of canister **200** measured in the longitudinal direction. In example aspects, a portion of second securing channel **214** may be positioned at any a location from about 0.25 in. (0.64 cm) to about 7.5 in (19.1 cm) away from second canister end **204**, as measured in the longitudinal direction.

(35) For those aspects that comprise at least two securing channels, securing channels may each be separated by a distance. As illustrated in FIG. 2, first securing channel **212** and second securing

channel **214** are separated by a distance measured in the longitudinal direction. While securing channels may be separated by any distance, in an example aspect, first securing channel **212** and second securing channel **214** are positioned at any location such that first securing channel **212** is separated from second securing channel **214** anywhere from about 1 in. (2.5 cm) to about 10 in. (25.4 cm).

(36) In some aspects, a distance from first securing channel **212** to first canister end **202** is less than a distance from first securing channel **212** to second canister end **204**. In some aspects, the distance from first securing channel **212** to first canister end **202** is less than a distance from first securing channel **212** to second securing channel **214**.

(37) FIG. 3A provides a cross sectional view of canister **200** of FIG. 2. The cross section is taken along the longitudinal direction at center aspect **210**. FIG. 3A illustrates first securing channel **212**. The cross sectional view illustrates first securing channel **212** and second securing channel **214** having extended around at least a portion of canister **200**.

(38) As illustrated in FIG. 3A, first securing channel **212** comprises first channel volume **216** and second channel volume **218**. Further, as canister **200** is an example canister having two securing channels, canister **200** further comprises second securing channel **214**, which comprises third channel volume **234** and fourth channel volume **236**. Each of these will be more fully described with reference to FIG. 3B and FIG. 3C. As illustrated in FIG. 3A, second channel volume **218** of first securing channel **212** is more interior to center aspect **210** than first channel volume **216**, while fourth channel volume **236** is more interior to center aspect **210** than third channel volume **234**. Put another way, a distance from center aspect **210** to second channel volume **218** is less than a distance from center aspect **210** to first channel volume **216**. Likewise, in this example, a distance from center aspect **210** to fourth channel volume **236** is less than a distance from center aspect **210** to third channel volume **234**.

(39) In the aspect illustrated first securing channel **212** and second securing channel **214** are each exterior to wall **206**. In some aspects, a portion of first securing channel **212** or second securing channel **214** is exterior to wall **206**. In a particular aspect of the technology, one or more securing channels are integrally formed by wall **206**. That is, one or more features of a securing channel may be integrally formed from wall **206**.

(40) To provide further illustration of this example, FIG. 3B illustrates a portion of canister **200** that comprises first securing channel **212**. The figure illustrates a cross sectional view of first securing channel **212**. Here, FIG. 3B provides further illustration of a cross section of the first channel volume **216** and second channel volume **218**, which are show delineated by a theoretical dotted line.

(41) First channel volume **216** may be defined by, at least in part, first channel surface **220** and second channel surface **222**. First channel surface **220** and second channel surface **222** may extend in a direction perpendicular to the longitudinal direction. In the aspect illustrated, first channel surface **220** extends at least partially parallel with second channel surface **222** in the direction perpendicular to the longitudinal direction. First channel surface **220** is spaced apart from second channel surface **222**, at least partially forming first channel volume **216**. The distance corresponds to a first width of first channel volume **216** as measured in the longitudinal direction. To provide some examples, the first width may be any width from about 0.10 in (0.25 cm) to about 1.0 in. (2.54 cm).

(42) Each of first channel surface **220** and second channel surface **222** extend in a direction away from center aspect **210**, as illustrated in FIG. 3A. Thus, first channel volume **216** may also be defined at least partially by opposite terminal edges of each of first channel surface **220** and second channel surface **222** as each extends away from center aspect **210**.

(43) Second channel volume **218** may be defined, at least in part, by third channel surface **224** and fourth channel surface **226**. In the example shown, third channel surface **224** is separated from fourth channel surface **226** by a distance that is about equal to the first width, e.g., the distance over

which first channel surface **220** and second channel surface **222** are separated. Third channel surface **224** and fourth channel surface **226** are illustrated as extending in the longitudinal direction. In the illustrated aspect, second channel volume **218** is further at least partially defined by fifth channel surface **228**, which extends in the longitudinal direction and is separated from each of third channel surface **224** and fourth channel surface **226** by a distance.

(44) Second channel volume **218** may further be defined, at least in part, by sixth channel surface **230** and seventh channel surface **232**. Sixth channel surface **230** and seventh channel surface **232**, as illustrated in this example, are parallel and extend in the direction perpendicular to the longitudinal direction. Each of sixth channel surface **230** and seventh channel surface **232** extend in a direction away from center aspect **210**, as illustrated in FIG. 3A. Second channel volume **218** may also be defined at least partially by opposite terminal edges of each of sixth channel surface **230** and seventh channel surface **232** as each extends away from center aspect **210**. In this example a first terminal edge of each of sixth channel surface **230** and seventh channel surface **232** intersects fifth channel surface **228**, while a second terminal edge of sixth channel surface **230** intersects third channel surface **224** and a second terminal edge of seventh channel surface **232** intersects fourth channel surface **226**.

(45) Sixth channel surface **230** and seventh channel surface **232** are spaced apart and separated by a distance. The distance over which sixth channel surface **230** and seventh channel surface **232** extends corresponds with a second width of second channel volume **218**. To provide some examples, the second width may be greater than about 0.10 in (0.25 cm) to about 1.5 in. (3.8 cm).

(46) To facilitate releasably coupling canister **200** to a delivery vehicle, e.g., using a rail bracket that will be further described, by way of first securing channel **212**, the first width is less than the second width. Put another way, the distance from first channel surface **220** to second channel surface **222** is less than the distance from sixth channel surface **230** to seventh channel surface **232**.

(47) FIG. 3C illustrates a portion of canister **200** that comprises second securing channel **214**. The figure illustrates a cross sectional view of second securing channel **214**. Here, FIG. 3C provides further illustration of a cross section of the third channel volume **234** and fourth channel volume **236**, which are show delineated by a theoretical dotted line.

(48) Third channel volume **234** may be defined by, at least in part, eighth channel surface **238** and ninth channel surface **240**. Eighth channel surface **238** and ninth channel surface **240** may extend in a direction perpendicular to the longitudinal direction. In the aspect illustrated, eighth channel surface **238** extends at least partially parallel with ninth channel surface **240** in the direction perpendicular to the longitudinal direction. Eighth channel surface **238** is spaced apart from ninth channel surface **240**, at least partially forming third channel volume **234**. The distance corresponds to a third width of third channel volume **234** as measured in the longitudinal direction. To provide some examples, the third width may be any with from about 0.10 in (0.25 cm) to about 1.0 in. (2.54 cm). In a particular aspect, the third width is the same as the first width, thereby allowing canister **200** to be coupled in any orientation.

(49) Each of eighth channel surface **238** and ninth channel surface **240** extend in a direction away from center aspect **210**, as illustrated in FIG. 3A. Thus, third channel volume **234** may also be defined at least partially by opposite terminal edges of each of eighth channel surface **238** and ninth channel surface **240** as each extends away from center aspect **210**.

(50) Fourth channel volume **236** may be defined, at least in part by tenth channel surface **242** and eleventh channel surface **244**. In the example shown, tenth channel surface **242** is separated from eleventh channel surface **244** by a distance that is about equal to the third width, e.g., the distance over which eighth channel surface **238** and ninth channel surface **240** are separated. Tenth channel surface **242** and eleventh channel surface **244** are illustrated as extending in the longitudinal direction. In the illustrated aspect, fourth channel volume **236** is further at least partially defined by twelfth channel surface **246**, which extends in the longitudinal direction and is separated from each of tenth channel surface **242** and eleventh channel surface **244** by a distance.

(51) Fourth channel volume **236** may further be defined, at least in part, by thirteenth channel surface **248** and fourteenth channel surface **250**. Thirteenth channel surface **248** and fourteenth channel surface **250**, as illustrated in this example, are parallel and extend in the direction perpendicular to the longitudinal direction. Each of thirteenth channel surface **248** and fourteenth channel surface **250** extend in a direction away from center aspect **210**, as illustrated in FIG. 3A. Fourth channel volume **236** may also be defined at least partially by opposite terminal edges of each of thirteenth channel surface **248** and fourteenth channel surface **250** as each extends away from center aspect **210**. In this example a first terminal edge of each of thirteenth channel surface **248** and fourteenth channel surface **250** intersects twelfth channel surface **246**, while a second terminal edge of thirteenth channel surface **248** intersects tenth channel surface **242** and a second terminal edge of fourteenth channel surface **250** intersects eleventh channel surface **244**.

(52) Thirteenth channel surface **248** and fourteenth channel surface **250** are spaced apart and separated by a distance. The distance over which thirteenth channel surface **248** and fourteenth channel surface **250** extends corresponds with a fourth width of fourth channel volume **236**. To provide some examples, the fourth width may be greater than about 0.10 in. (0.25 cm) to about 1.5 in. (3.8 cm). In a particular aspect, the second width is the same as the fourth width, thereby allowing canister **200** to be coupled in any orientation.

(53) To facilitate releasably coupling canister **200** to an object by way of second securing channel **214**, the third width can be less than the fourth width. Put another way, the distance from eighth channel surface **238** to ninth channel surface **240** is less than the distance from thirteenth channel surface **248** to fourteenth channel surface **250**.

(54) Each of the channel surfaces described with reference to FIG. 3B and FIG. 3C may extend entirely or at least partially around canister **200**. In an aspect, such as the one illustrated by FIGS. 3A-3C, first securing channel **212** extends parallel with at least a portion of second securing channel **214**.

(55) Turning now to FIG. 4, an example rail bracket **300** is provided. Rail bracket **300** is one example that is suitable for use in releasably coupling canister **200** via first securing channel **212** and second securing channel **214**.

(56) In this example, rail bracket **300** comprises first bracket surface **302** and second bracket surface **304**. First bracket surface **302** is opposite and spaced apart from second bracket surface **304**. Rail bracket **300** extends between first bracket surface **302** and second bracket surface **304**. Rail bracket **300** comprises rail bracket length **306** in a first direction and rail bracket width **308** in second direction. Rail bracket length **306** and rail bracket width **308** may be measured anywhere at or between first bracket surface **302** and second bracket surface **304**. Rail bracket length **306** may be greater than rail bracket width **308**. In an aspect, rail bracket **300** tapers inward from first bracket surface **302** to second bracket surface **304**.

(57) In general, rail brackets may comprise one or more rotational shafts. In the example provided by FIG. 4, rail bracket **300** comprises at least first rotational shaft **310**. First rotational shaft **310** comprises first shaft end **312** that is opposite from second shaft end **314**. First rotational shaft **310** extends longitudinally between first shaft end **312** and second shaft end **314** and has a first longitudinal length measured between first shaft end **312** and second shaft end **314**. The longitudinal length may be greater than a distance measured between first bracket surface **302** and second bracket surface **304**. As such, first rotational shaft **310** can extend from first shaft end **312** through first bracket surface **302** and second bracket surface **304** to second shaft end **314**.

(58) First rotational shaft **310** comprises first rotational axis **316**. First rotational axis **316** is represented by way of a theoretical dashed line. First rotational axis **316** may extend through first rotational shaft **310** along the first longitudinal length, and theoretically extend through first shaft end **312** and second shaft end **314**. As such, at least a portion of first rotational shaft **310** extends a first radial distance outward from first rotational axis **316**. First rotational shaft **310** may rotate about, at least partially or fully, first rotational axis **316**. In a particular aspect, first rotational shaft

310 rotates about first rotational axis **316** by at least 90° (degrees). While first rotational shaft **310** is illustrated as a cylindrical object, nothing is intended to limit first rotational shaft **310** to only this example, and other geometric configurations are contemplated as also suitable.

(59) Rail bracket **300** may comprise one or more fingers that extend from first rotational shaft **310** beyond the first radial distance. In the aspect illustrated by FIG. 4, first rotational shaft **310** comprises first finger **318** proximate second shaft end **314**. That is, first finger **318** may be positioned at second shaft end **314** or at a location along first rotational shaft **310**, such that the location at which first finger **318** is positioned is closer to second shaft end **314** than first shaft end **312**. First finger **318** may be a separate component or integrally formed as part of first rotational shaft **310**. In general, first finger **318** extends away from first rotational axis **316** over a second radial distance, and the second radial distance is greater than the first radial distance. In the example provided, first finger **318** comprises a length measured in a direction extending away from first rotational axis **316** that is greater than a width of first finger **318**.

(60) As noted, first rotational shaft **310** rotates about first rotational axis **316**. In this way, first finger **318** can be transitioned between a first position and a second position. In the example illustrated, in the first position, first finger **318** extends away from first rotational axis **316** along the first direction defined by rail bracket length **306**. When first rotational shaft **310** rotates, first finger **318** is transitioned to a second position that is offset from the first position based on the degree of rotation. In a particular case, first rotational shaft **310** rotates about 90°, thereby causing the second position to be offset from the first position by the same about 90°.

(61) In the particular example provided by FIG. 4, rail bracket **300** comprises second finger **320** that extends from first rotational shaft **310** in a direction opposite that of first finger **318**. Second finger **320** may be a separate component or be an integral component of first rotational shaft **310** or first finger **318**. As illustrated, second finger **320** is proximate second shaft end **314**. That is, second finger **320** may be positioned at second shaft end **314** or at a location along first rotational shaft **310**, such that the location at which second finger **320** is positioned is closer to second shaft end **314** than first shaft end **312**. Second finger **320** extends away from first rotational axis **316** a third radial distance that is greater than the first radial distance. In the example provided, second finger **320** comprises a length measured in a direction extending away from first rotational axis **316** that is greater than a width of second finger **320**. When first rotational shaft **310** rotates, second finger **320** may similarly transition from a first position to a second position that is offset from the first position based on the degree of rotation. In a particular case, first rotational shaft **310** rotates about 90°, thereby causing the second position to be offset from the first position by the same about 90°. In the first position, second finger **320** may extend away from first rotational axis **316** in a direction that corresponds to the first direction defined by rail bracket length **306**.

(62) As described, rail bracket **300** may comprises one or more rotational shafts, including second rotational shaft **322**. Second rotational shaft **322** comprises third shaft end **324** that is opposite from fourth shaft end **326**. Second rotational shaft **322** extends longitudinally between third shaft end **324** and fourth shaft end **326** and has a second longitudinal length measured between third shaft end **324** and fourth shaft end **326**. The longitudinal length may be greater than a distance measured between first bracket surface **302** and second bracket surface **304**. As such, second rotational shaft **322** can extend from third shaft end **324** through first bracket surface **302** and second bracket surface **304** to fourth shaft end **326**.

(63) Second rotational shaft **322** comprises second rotational axis **328**. Second rotational axis **328** is represented by way of a theoretical dashed line. Second rotational axis **328** may extend through second rotational shaft **322** along the second longitudinal length, and theoretically extend through third shaft end **324** and fourth shaft end **326**. As such, at least a portion of second rotational shaft **322** extends a fourth radial distance outward from second rotational axis **328**. Second rotational shaft **322** may rotate about, at least partially or fully, second rotational axis **328**. In a particular aspect, second rotational shaft **322** rotates about second rotational axis **328** by at least 90°. While

second rotational shaft **322** is illustrated as a cylindrical object, nothing is intended to limit second rotational shaft **322** to only this example, and other geometric configurations are contemplated as also suitable.

(64) Rail bracket **300** may comprise one or more fingers that extend from second rotational shaft **322** beyond the fourth radial distance. In the aspect illustrated by FIG. 4, second rotational shaft **322** comprises third finger **330** proximate fourth shaft end **326**. That is, third finger **330** may be positioned at fourth shaft end **326** or at a location along second rotational shaft **322**, such that the location at which third finger **330** is positioned is closer to fourth shaft end **326** than third shaft end **324**. Third finger **330** may be a separate component or integrally formed as part of second rotational shaft **322**. In general, third finger **330** extends away from second rotational axis **328** over a fifth radial distance, and the fifth radial distance is greater than the fourth radial distance. In the example provided, third finger **330** comprises a length measured in a direction extending away from second rotational axis **328** that is greater than a width of third finger **330**.

(65) As noted, second rotational shaft **322** rotates about second rotational axis **328**. In this way, third finger **330** can be transitioned between a third position and a fourth position. In the example illustrated, in the third position, third finger **330** extends away from second rotational shaft **322** along the first direction defined by rail bracket length **306**. When second rotational shaft **322** rotates, third finger **330** is transitioned to a fourth position that is offset from the third position based on the degree of rotation. In a particular case, second rotational shaft **322** rotates about 90°, thereby causing the fourth position to be offset from the third position by the same about 90°.

(66) In the particular example provided by FIG. 4, rail bracket **300** comprises fourth finger **332** that extends from second rotational shaft **322** in a direction opposite that of third finger **330**. Fourth shaft end **326** may be a separate component or be an integral component of second rotational shaft **322** or third finger **330**. As illustrated, fourth finger **332** is proximate fourth shaft end **326**. That is, fourth finger **332** may be positioned at fourth shaft end **326** or at a location along second rotational shaft **322**, such that the location at which fourth finger **332** is positioned is closer to fourth shaft end **326** than third shaft end **324**. Fourth finger **332** extends away from second rotational axis **328** a sixth radial distance that is greater than the fourth radial distance. In the example provided, fourth finger **332** comprises a length measured in a direction extending away from second rotational axis **328** that is greater than a width of fourth finger **332**. When second rotational shaft **322** rotates, fourth finger **332** may similarly transition from a third position to a fourth position that is offset from the third position based on the degree of rotation. In a particular case, second rotational shaft **322** rotates about 90°, thereby causing the fourth position to be offset from the third position by the same about 90°. In the third position, fourth finger **332** may extend away from second rotational axis **328** in a direction that corresponds to the first direction defined by rail bracket length **306**.

(67) In some aspects, a second bracket surface is shaped to correspond with a container with which the rail bracket will releasably couple. That is, one portion of the second bracket surface may have a distance from the first bracket surface that is different from a distance between a second portion of the second bracket surface and the first bracket surface. Looking at FIG. 4, second bracket surface **304** comprises an arc that is used to enhance coupling to a canister, such as canister **200** of FIG. 2. With reference still to FIG. 4, first portion **334** of second bracket surface **304** has a distance from first bracket surface **302** that is less than a distance measured from second portion **336** of second bracket surface **304** to first bracket surface **302**. In this case, second bracket surface **304** curves outward and away from first bracket surface **302** in an outward direction from a center aspect of second bracket surface **304**.

(68) Moreover, in some aspects that comprises more than one rotational shaft, the rotational shafts may be angled based on an object with which the rail bracket will releasably couple. As noted, one suitable use of rail bracket **300** is to releasably couple a substantially cylindrical canister. As such, first rotational shaft **310** and second rotational shaft **322** are angled such that respective second shaft end **314** and fourth shaft end **326** engage a securing channel of the canister. In this particular

example, first rotational axis **316** and second rotational axis **328** theoretically extend beyond second shaft end **314** and fourth shaft end **326**, respectively, and are convergent in a direction extending outward and away from second bracket surface **304**. It will be understood that rotational shafts in other aspects may have the same angle or be offset by any degree to correspond to the container size or shape.

(69) In an aspect of the technology, the one or more rotational shafts are biased in a direction toward a first shaft end. Such bias may be provided by a spring or other mechanical feature applying a force to the one or more rotational shafts in the direction toward the first shaft end. Using the example provided by FIG. 4, first rotational shaft **310** or second rotational shaft **322** may be biased in a direction from second shaft end **314** toward first shaft end **312** and fourth shaft end **326** toward third shaft end **324**, respectively. Such features may provide additional locking benefits that will be further described.

(70) As will be further discussed, rail bracket **300** may be employed with a delivery vehicle, such as a UAV, to retrieve an object for delivery, such as a canister corresponding to those described. FIG. 1 illustrates one such example. When employed, first bracket surface **302** can be exposed to the delivery vehicle, such as the UAV. That is, first bracket surface **302** is closer to the delivery vehicle relative to second bracket surface **304** when rail bracket **300** is secured to the delivery vehicle, either directly or indirectly.

(71) Various devices may be employed to rotate rotational shafts, such as first rotational shaft **310** and second rotational shaft **322**. These include electric motors, actuators, servos, solenoids, and the like. One particular device suitable for use is a rotary solenoid, some of which may be configured to rotate between two positions by a predetermined or configurable degree. When employed in use with a delivery vehicle, the delivery vehicle can provide an electrical signal to the rotary solenoid, thereby causing it to rotate a rotational shaft between a first and second position. When used in coordination with a UAV, the rotary solenoid may be in communication with the flight controller, providing control over the rotational shaft and fingers that are used to releasably couple an object for delivery, such as a canister, thereby allowing the UAV to retrieve and release the object.

(72) FIG. 5A and FIG. 5B are provided to illustrate an example of using rail bracket **300** to releasably couple canister **200** by way of first securing channel **212**.

(73) FIG. 5A is a cross sectional view of a portion of rail bracket **300** as at least partially inserted into first securing channel **212** of canister **200**. Thus, rail bracket **300** is inserted into at least a portion of first securing channel **212**. During insertion, first finger **318** is in a first position such that it is inserted through first width of first channel volume **216**, thereby allowing first finger **318** to be disposed within second channel volume **218**.

(74) FIG. 5B is a cross sectional view of a portion of rail bracket **300** as inserted into first securing channel **212** of canister **200**. As illustrated, first rotational shaft **310** is rotated about first rotational axis **316**. In doing so, first finger **318** is rotated within second channel volume **218** and transitioned to a second position. In the example provided by FIG. 5B, first rotational shaft **310** has been rotated 90° and second finger **320** is now in view. Here, the relative positions of first finger **318** and second finger **320** are such that a relative length measured from along first finger **318** and second finger **320** is greater than the first width of first channel volume **216**, thereby securing second shaft end **314** within second channel volume **218** and coupling canister **200** with rail bracket **300**. By rotation of first rotational shaft **310** between the positions illustrated by FIGS. 5A and 5B, rail bracket **300** couples and decouples with canister **200**. When used with a delivery vehicle, such as a UAV, the rail bracket provides a mechanism by which the delivery vehicle can retrieve and release the object being delivered.

(75) It will be appreciated that other like components described herein may operate in a similar manner.

(76) Turning now to FIG. 6, an example UAV **400** is provided having example canister **402** coupled thereto. Coupling of canister **402** to UAV **400** is performed using a rail bracket (not visible in FIG.

6), such as rail bracket **300**. Moreover, canister **200** is one example suitable for use as canister **402**. (77) In this example, when coupled, canister **402** rotates about the rail bracket such that a center of gravity of the canister is aligned with a center of gravity of the UAV. This is due, at least in part, to a securing channel at least partially extending around canister **402**. As such, components of the rail bracket slide through the securing channel as canister **402** rotates due to gravitational force. The rail bracket of UAV **400** is positioned along the center of gravity of UAV **400**. As such, when canister **402** rotates after coupling, as indicated by rotational arrow **404**, the center of gravity of canister **402** aligns with the center of gravity of UAV **400**. By aligning the center of gravities, UAV **400** is more efficient during some flight patterns.

(78) When UAV **400** has moved from a first location to a second location, canister **402** can be decoupled from UAV **400** by way of the rail bracket as previously described. As illustrated in FIG. 7, upon decoupling canister **402**, canister **402** is released from UAV **400** at the second location, as indicated by arrow **406**.

(79) With reference now to FIG. 8, a side perspective, cross sectional view of another aspect of the technology is provided. Illustrated is securing channel **500**. As similarly described, securing channel **500** comprises first channel volume **502** and second channel volume **504**. First channel volume **502** is defined at least in part by first channel surface **506** and second channel surface **508**, while second channel volume **504** is defined at least in part by third channel surface **510** and fourth channel surface **512**.

(80) In general, a securing channel may comprises one or more grooves on a channel surface of the securing channel. In the example aspect illustrated, securing channel **500** comprises one or more grooves disposed at third channel surface **510** and fourth channel surface **512**. Of the one or more grooves, **514a-d** are can be seen in the illustration. The one or more grooves can extend along third channel surface **510** and fourth channel surface **512** substantially perpendicular to first channel surface **506** and second channel surface **508**.

(81) In aspects, groove size may correspond to a finger width of a rail bracket, such as any of those described herein. The aspect provided by securing channel **500**, may be used with any of the canisters described herein, and to facilitate coupling the canister via the rail bracket to a delivery vehicle. When used, securing channel **500** allows the canister to lock into a certain orientation. The finger of a rail bracket nests within a groove of the one or more grooves and inhibits rotation of the canister. In cases where the rail bracket comprises a biased rotational shaft, the bias helps maintain the position of the finger within the groove. This aspects is beneficial when rotation of the canister is not desired or when a particular orientation of the canister is desired.

(82) The subject matter of the present technology is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed or disclosed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” or “block” might be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly stated.

(83) Embodiments described above may be combined with one or more of the specifically described alternatives. In particular, an embodiment that is claimed may contain a reference, in the alternative, to more than one other embodiment. The embodiment that is claimed may specify a further limitation of the subject matter claimed.

(84) In addition, words such as “a” and “an,” unless otherwise indicated to the contrary, include the plural as well as the singular. Thus, for example, the constraint of “a feature” is satisfied where one or more features are present. Furthermore, the term “or” includes the conjunctive, the disjunctive, and both (a or b thus includes either a or b, as well as a and b).

(85) Terms, such as “having,” “including,” or the like, and derivatives thereof, are intended to have the same broad meaning as “comprising,” unless otherwise expressly indicated.

(86) Unless explicitly indicated otherwise, ranges are intended to provide example embodiments that are suitable for some aspects and implementations. They are not intended to exclude other embodiments of the technology having values falling outside of the given ranges.

(87) From the foregoing, it will be seen that this technology is one well adapted to attain all the ends and objects described above, including other advantages that are obvious or inherent to the structure. It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is within the scope of the claims. Since many possible embodiments of the described technology may be made without departing from the scope, it is to be understood that all aspects described herein or illustrated the accompanying drawings is to be interpreted as illustrative and not in a limiting sense.

Claims

1. A canister comprising: a first canister end opposite a second canister end defining a longitudinal direction extending between the first canister end and the second canister end; a wall extending between the first canister end and the second canister end and defining at least in part an interior volume of the canister; and a securing channel having a first channel volume and a second channel volume, the first channel volume having a first width in the longitudinal direction, and the second channel volume having a second width in the longitudinal direction and comprising a first channel surface and a second channel surface, wherein the first width is less than the second width, the second channel volume is more interior to a center aspect of the interior volume than the first channel volume, and the first channel surface and the second channel surface comprise a groove to allow a finger of a rail bracket to nest within the groove to couple the rail bracket to the securing channel.
2. The canister of claim 1, wherein the groove is configured to inhibit rotation of the canister when the finger is nested within the groove.
3. The canister of claim 1, wherein the first channel volume comprises a third channel surface and a fourth channel surface, and the groove extends along the first channel surface and the second channel surface substantially perpendicular to the third channel surface and the fourth channel surface.
4. The canister of claim 1, wherein a size of the groove corresponds to a width of the finger.
5. The canister of claim 1, further comprising a second securing channel having a third channel volume and a fourth channel volume, the third channel volume having a third width in the longitudinal direction, and the fourth channel volume having a fourth width in the longitudinal direction and comprising a third channel surface and a fourth channel surface, wherein the third width is less than the fourth width, the fourth channel volume is more interior to the center aspect of the interior volume than the third channel volume, the third channel surface and the fourth channel surface comprise a second groove to allow a finger of a second rail bracket to nest within the second groove to couple the second rail bracket to the second securing channel, and the securing channel is separated from the second securing channel by a distance in the longitudinal direction.
6. The canister of claim 5, wherein the securing channel extends parallel with at least a portion of the second securing channel.
7. The canister of claim 5, wherein a second distance from the securing channel to the first canister end is less than the distance separating the securing channel and the second securing channel.
8. A canister comprising: a first canister end opposite a second canister end defining a longitudinal direction extending between the first canister end and the second canister end; a wall extending between the first canister end and the second canister end and defining at least in part an interior

volume of the canister; and a securing channel having a first channel volume and a second channel volume, the first channel volume having a first width in the longitudinal direction, and the second channel volume having a second width in the longitudinal direction and comprising a first channel surface and a second channel surface, wherein the first channel surface and the second channel surface comprise a groove to allow a finger of a rail bracket to nest within the groove to couple the rail bracket to the securing channel.

9. The canister of claim 8, wherein the groove is configured to inhibit rotation of the canister when the finger is nested within the groove.

10. The canister of claim 8, wherein the first channel volume comprises a third channel surface and a fourth channel surface, and the groove extends along the first channel surface and the second channel surface substantially perpendicular to the third channel surface and the fourth channel surface.

11. The canister of claim 8, wherein a size of the groove corresponds to a width of the finger.

12. The canister of claim 8, further comprising a second securing channel having a third channel volume and a fourth channel volume, the third channel volume having a third width in the longitudinal direction, and the fourth channel volume having a fourth width in the longitudinal direction and comprising a third channel surface and a fourth channel surface, wherein the third channel surface and the fourth channel surface comprise a second groove to allow a finger of a second rail bracket to nest within the second groove to couple the second rail bracket to the second securing channel, and the securing channel is separated from the second securing channel by a distance in the longitudinal direction.

13. The canister of claim 12, wherein the securing channel extends parallel with at least a portion of the second securing channel.

14. The canister of claim 12, wherein a second distance from the securing channel to the first canister end is less than the distance separating the securing channel and the second securing channel.

15. A method comprising: inserting a rail bracket within at least a portion of a securing channel of a canister, wherein: the rail bracket comprises: a first bracket surface exposed to a vehicle, a rotational shaft having a first shaft end and a second shaft end with a longitudinal length extending between the first shaft end and the second shaft end, the rotational shaft extending through the first bracket surface and a first radial distance from a rotational axis extending along the longitudinal length, and a finger extending radially from the rotational shaft proximate the second shaft end, the finger having a second radial distance from the rotational axis that is greater than the first radial distance, and the securing channel has a first channel volume and a second channel volume, the first channel volume having a first width in a longitudinal direction of the canister, and the second channel volume having a second width in the longitudinal direction of the canister, and the second channel volume comprising a first channel surface and a second channel surface that comprise a groove; and rotating the rotational shaft such that the finger transitions between a first position perpendicular to the longitudinal direction of the canister and a second position offset from the first position to allow the finger to nest within the groove.

16. The method of claim 15, wherein when the finger is in the first position, the canister is decoupled from the vehicle, and when the finger is in the second position, the canister is coupled to the vehicle.

17. The method of claim 15, wherein when the finger is in the second position, the canister is inhibited from rotating.

18. The method of claim 15, wherein the second position is offset from the first position by about ninety degrees.

19. The method of claim 15, wherein the vehicle is an unmanned aerial vehicle (UAV).

20. The method of claim 15, wherein the vehicle provides an electrical signal to a rotary solenoid,

thereby rotating the rotational shaft such that the finger transitions between the first position and the second position.
